Slow to Hire, Slow to Fire: Labour Adjustment Frictions in French Viticulture

Fabien CANDAU[*,†]

Laure LATRUFFE[‡,§]

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Abstract

To what extent are French wine farms slow to adjust their workforce to demand fluctuations? In this paper, we measure the cost of adjusting variable labour using the 2003–2024 French Farm Accountancy Data Network (FADN) panel. Using a quasi-natural experiment, we document that variable labour barely moves when sales collapse. We identify the same sluggishness on the upside with instrumental variables built from foreign-market variation. A one-standard-deviation increase in demand raises variable-labour hiring by about 33 percent, about half of what a farm would add in the absence of any adjustment costs. A structural model with heterogeneous farm productivity and permanent labour acting as a production buffer translates this elasticity into a per-worker contraction cost. Removing the friction would raise the average farm's operating profit by about 0.9% per year (about €54 million annually for the sector). The gain is less about cutting labour in downturns than about the hiring farms forgo in good years for fear of costly future contraction. The cost concentrates on farms in the upper half of the productivity distribution.

[*]Univ. Bordeaux, CNRS, BxSE, UMR 6060, UMR 1441, INRAE, F-33600 Pessac, France; and UPPA, CNRS, TREE. Email: fabien.candau@univ-pau.fr.

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[‡]Univ. Bordeaux, CNRS, BxSE, UMR 6060, UMR 1441, INRAE, F-33600 Pessac, France.

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1 Introduction

French wine is a sector of persistent sales shocks.[1] Farms can meet these swings by expanding or shrinking the workforce, but that adjustment is costly. The labour-adjustment-cost literature has measured that cost at the plant level in manufacturing, but its magnitude in agriculture, and its incidence across heterogeneous farms, have been hard to measure. Adjustment costs have several sources. They can stem from the specific knowledge a worker builds up on the job, from training and coordination, or from institutional features of the labour market such as severance pay and end-of-contract costs.

We do not identify which of these sources is responsible for the adjustment costs; instead, in this paper, we measure them directly, using the French Farm Accountancy Data Network (FADN) panel of 1,720 wine farms over 2003–2024.[2] Adjustment at the farm level happens primarily on variable labour, defined as the non-permanent component of the workforce, including fixed-term contracts renewed across years and seasonal contracts. Two findings on variable labour motivate the analysis. About a quarter of farm-years operate without any variable labour at all. In the other three-quarters, variable labour moves with sales, but only about half as much as it would if farms could adjust their workforce freely.

A quasi-natural experiment motivates the analysis and shows the friction at work. When a climate catastrophe makes a farm's sales collapse, variable labour barely responds, even as the shortfall persists for years. The same friction shapes hiring when sales boom. A forward-looking farm hires less than demand warrants, knowing that workers added today may be costly to shed tomorrow. To identify this response we need changes in the demand for French wine that the farm did not cause. The weather of a large competitor provides them. The United States is a major wine producer and importer, so a poor season in California ripples through the world wine trade and lifts the demand facing French producers in the markets where they compete. That Californian weather reaches a French farm's labour only through the demand for its wine, never directly. We use the elasticity estimated by this IV strategy as a central input to a structural model à la Cooper and Haltiwanger (2006). We find that the demand increase raises variable-labour hiring well below what an unconstrained farm would add.

Behind this rigidity is a per-worker cost of contracting variable labour. We recover it via indirect inference on an adjustment-cost model with two features that

[1] Since 2008, the industry has absorbed the financial crisis, a sharp increase and then decrease in Chinese demand, Brexit and U.S. protectionism, stop-and-go disruptions due to COVID, the Russia–Ukraine war, and a run of adverse harvests.

[2] Survey-weighted; full sample description in Section 4.1.

fit French viticulture. First, permanent labour produces output even when no variable labour is hired, so a farm can choose to operate with permanent staff alone in low-revenue years. Second, farms differ in a permanent productivity dimension, which we calibrate from the dispersion of farm fixed effects in the production function. The model matches the IV-identified elasticity and the share of farm-years operating with permanent staff alone. From that model, we find that removing the adjustment cost would raise operating profit by about 0.9% per year on average across the population, on the order of €54 million annually for the French wine sector, holding input and output prices fixed. The cost concentrates on farms in the middle and upper half of the productivity distribution. Median farms pay for keeping variable labour they would otherwise let drop to zero in low-revenue years, while mid-large farms under-hire when demand is high and bear roughly two-thirds of the aggregate loss.

The structural adjustment-cost framework we apply originates with Cooper and Haltiwanger (2006), Cooper and Willis (2009), and Bloom (2009). These papers estimate the parameters of adjustment-cost models, on capital or labour or both, using plant-level manufacturing panels. While agricultural economics has a long literature on farm labour,[3] the structural adjustment-cost framework we apply has not been brought to that setting. We extend the framework in two ways. First, we apply it at the farm level in agriculture; the climate-driven volatility of farm revenue and the option to operate with permanent staff alone are features the manufacturing applications do not face. Second, we recover the auxiliary moment causally, whereas these papers recover it by OLS. This brings to the adjustment-cost literature the causal-IV-into-structural turn that Berry and Compiani (2023) have recently pursued in industrial organisation. Weather-based instruments have a precedent in Roberts and Schlenker (2013), who exploit a crop's own yield shocks as a supply shifter to estimate world commodity elasticities. Our design departs from theirs on three counts. The instrument is a foreign competitor's weather, not the home crop's; it shifts demand through the trade network rather than domestic supply; and the elasticity it identifies, recovered from differential exposure across French regions, feeds a structural labour model rather than a reduced-form market equilibrium.

The paper is organised as follows. Section 2 reviews the sources of labour adjustment cost and presents the quasi-natural experiment that motivates the model. Section 3 sets up the CES demand, the Cobb–Douglas production technology, and the dynamic-friction model that the empirical strategy targets. Section 4 describes

[3] On farmers' labour supply, see Sumner (1982) and Rutledge and Mérel (2023); on migration and the farm workforce, Charlton and Taylor (2016) and Clemens et al. (2018); on the family-versus-hired-labour distinction, Benjamin (1992) and LaFave and Thomas (2016).

the data and the instrumental variable strategy. Section 5 reports the IV estimates. Section 6 closes the model, runs the indirect inference, and quantifies the operating-profit cost of the labour friction across farm types. Section 7 concludes.

2 Background

This section reviews the sources of labour adjustment cost and documents the friction in a quasi-natural experiment.

2.1 About adjustment costs

A basic assumption in microeconomic textbooks is that labour adjusts instantaneously to minimise costs. In practice, a long literature beginning with Clark (1923) and Oi (1962) has emphasised that labour is often quasi-fixed (e.g., Bouchet et al., 1989; Moschini, 1989). In viticulture, quasi-fixity is arguably even stronger, because grapevines are perennial, many tasks require plot-specific knowledge, and production under quality schemes adds rigidity through rules on practices and yields. For organic production, these frictions are amplified further through pest management without synthetic inputs, compliance requirements, and organisational adaptation (Carneiro et al., 2024).

The literature distinguishes three broad classes of adjustment cost, each grounded in a distinct micro-foundation.

First, adjustment costs may be convex. The dynamic labour-demand tradition of Sargent (1978) models them as quadratic in the change of employment, with internal training, ramp-up, and coordination costs as the canonical micro-foundation. Viticulture displays several of these channels. Many parcels have their own soil profile, slope, microclimate, and rootstock-clone composition; even when general viticulture training is provided by vocational schools, the parcel-specific component is by definition farm-specific and accumulates on the job. The same gradual ramp-up applies across the wine sector, from vineyard work and treatment timing to cellar and lab procedures, and is more pronounced on organic or biodynamic estates, which add mechanical-weeding and biocontrol skills to the standard apprenticeship. To the extent that the farm's training resources (experienced staff time, supervisor attention, on-the-job practice slots) are not freely expandable in the short run, the marginal training cost tends to rise with the size of the hiring round, generating the convex shape.

Second, adjustment costs may be non-convex. The non-convex employment-adjustment literature (Hamermesh, 1989; Caballero et al., 1997; Cooper and Willis,

2009) treats adjustment as a discrete event that triggers a fixed cost, capturing indivisibilities, integer constraints, and contract-negotiation overhead. Viticulture displays several of these features. A vineyard's pruning team is hired and coordinated as a unit, and a salaried position is either filled or not. Each hiring round also carries recruitment overhead (transport, lodging, paperwork, and supervisor setup) with some components that do not scale linearly with team size. To the extent that these fixed components and indivisibilities are not negligible, they imply that producers absorb small mismatches and adjust only in discrete, lumpy episodes.

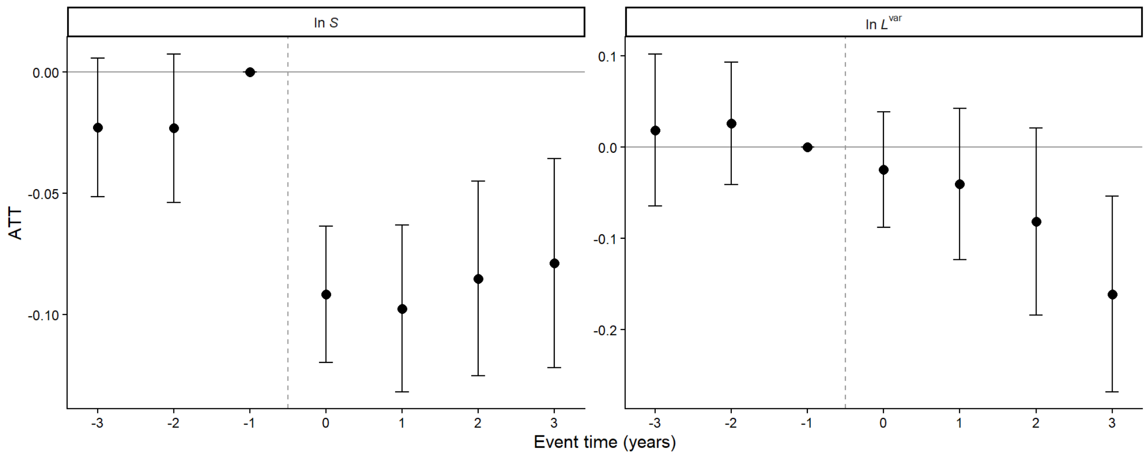
Third, adjustment costs may be asymmetric. The dynamic labour-demand tradition of Bentolila and Bertola (1990) and Bertola (1992) treats adjustment as costly only on the contraction side. These adjustment costs may be linked to the objective of retaining a qualified workforce through downturns in anticipation of recovery. They can also arise from institutional features of the labour market (severance indemnities, the legal cost of dismissal, end-of-contract bonuses, hiring search frictions), though the asymmetric-contraction-cost concept is generic rather than specific to any one regulatory regime.

The three forms are not mutually exclusive, and all three plausibly operate in viticulture. We adopt the asymmetric form on the variable-labour margin for two reasons. First, the empirical pattern documented in the next subsection (a sharp, persistent revenue fall met with little labour response) is consistent with a friction biting on the contraction side. A symmetric convex cost would slow the response in both directions, and a pure fixed cost would produce inaction interrupted by lumpy jumps, a pattern the data do not display. Second, the institutional features highlighted above are widely understood to be substantial on the French labour market, making the contraction margin the most obvious channel for the friction.

The asymmetric form should not be read as a purely institutional friction. It also accommodates the dynamic implications of sunk capacity investments specific to the farm's seasonal crew. Many wine producers have made non-trivial fixed investments to lodge, transport, and equip their returning teams — in some cases the conversion of a barn or the acquisition of a small hotel to house seasonal workers, in others the purchase of picking buckets, transport bins, sorting tables, or sanitary installations sized for a target crew. Once such assets are in place, the per-worker amortisation rises as the crew shrinks. This is not a classical fixed adjustment cost — the cost is paid up front and its dynamic manifestation is smooth rather than lumpy — but its empirical signature, reluctance to contract, is indistinguishable from the asymmetric friction the model specifies. It is one of several economic channels that the reduced-form contraction friction aggregates.

2.2 A quasi-natural experiment: how variable labour responds to a sharp revenue fall

To illustrate adjustment costs in periods of falling revenue, we study how French wine farms respond to a climate catastrophe (a frost, hail, drought, or storm that damages the vineyard or cellar stock). We identify these events from insurance indemnities, recorded at the farm-year level, and treat the year of each farm's largest payout as the shock, keeping the 918 farms whose largest payout exceeds €10,000. Because the catastrophes strike different farms in different years, we estimate the dynamic response with the interaction-weighted event-study estimator of Sun and Abraham (2021), which is robust to the bias that two-way fixed effects suffer under staggered timing and heterogeneous effects (Goodman-Bacon, 2021). We use farm and year fixed effects and $h = -1$ as the reference period (construction and robustness in Appendix C). Identification rests on the catastrophe being unforeseeable when the farm commits its labour (so the shock predates the labour choice) and on treated and not-yet-treated farms following parallel paths absent the event, which we verify with the pre-event coefficients. If contracting labour is costly, we expect revenue to fall on impact while variable labour barely adjusts, even as the shortfall persists.



Note: Sun–Abraham IW event-study coefficients on log sales ($\ln S$, left panel) and log variable labour ($\ln L^{\text{var}}$, right panel), by event time $h = t - g_i$ where g_i is the year of farm i's largest indemnity payout. Cohort restricted to farms with maximum indemnity $\geq$ €10,000 ($N = 918$ wine farms). Vertical bars show 95% confidence intervals (farm-clustered). Reference period $h = -1$.

Figure 1: Event-study response on the indemnity-amount cohort

The two response paths are revealing (Figure 1). Revenue drops sharply at the event year, by about -0.09 in logs, and stays significantly depressed for three years. Variable labour, by contrast, barely moves. The event-year coefficient, about -0.025, is statistically indistinguishable from zero, and the response stays small and insignificant through the second post-event year. A larger coefficient of about -0.16 appears at the third year, but it is not robust. As we raise the indemnity threshold that defines the cohort, the estimate shrinks toward zero and loses significance (Appendix C). Pre-event coefficients are jointly insignificant, supporting parallel pre-trends. The picture is one of a sharp, persistent fall in revenue met with little detectable labour contraction at any horizon.

This asymmetry (a large, persistent revenue shock met with little labour response) is what the next subsection formalises as a cost that bites on the contraction margin, whose strength we identify from an independent source of revenue variation.

3 The Model

3.1 The Demand

Consumers maximise a constant-elasticity-of-substitution (CES) utility function over differentiated wine varieties. In each year t the consumer chooses quantities $\{q_{it}\}_{i\in\mathcal{I}_t}$ of varieties produced by farms in the set $\mathcal{I}_t$, maximising

$$U_t \;=\; \left[\sum_{i\in\mathcal{I}_t} \left(b_{it}\, q_{it}\right)^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)}, \tag{1}$$

subject to the budget constraint

$$\sum_{i\in\mathcal{I}_t} p_{it}\, q_{it} \;=\; E_t, \tag{2}$$

where p_{it} is the price of variety i in year t, E_t is total expenditure on wine in year t, $b_{it} > 0$ is a variety-specific taste shifter (capturing terroir reputation, brand recognition, perceived quality, marketing effort, and other persistent or transitory demand-side factors), and $\sigma > 1$ is the elasticity of substitution between varieties.

Solving the consumer's problem yields the demand function faced by each farm,

$$q^D_{it} \;=\; b_{it}^{\sigma-1} \left(p_{it}/P_t\right)^{-\sigma} \left(E_t/P_t\right), \tag{3}$$

where P_t is the CES price index over wine varieties:

$$P_t \;=\; \left[\sum_{i\in\mathcal{I}_t} b_{it}^{\sigma-1}\, p_{it}^{1-\sigma}\right]^{1/(1-\sigma)}. \tag{4}$$

The elasticity of substitution σ governs how willing consumers are to substitute between varieties when relative prices change.

3.2 The Supply

Each farm i produces its wine variety using labour and capital with a Cobb–Douglas technology:

$$Y_{it} = e^{z_i} \left(L_i^{\text{fix}} + L_{it}^{\text{var}}\right)^{\tilde{\alpha}_L} K_{it}^{\tilde{\alpha}_K}, \tag{5}$$

where L_{it}^{var} is variable labour (the firm's choice variable), K_{it} is productive capital, L_i^{fix} is the farm's permanent labour endowment (family workers and full-time employees on open-ended contracts), $\tilde{\alpha}_L$ and $\tilde{\alpha}_K$ are the technological production-function elasticities, and z_i is a permanent productivity shifter (technological efficiency).

The permanent labour input L_i^{fix} enters the labour aggregator additively. Output remains strictly positive when $L_{it}^{\text{var}} = 0$, since the permanent workforce alone produces $e^{z_i} \, (L_i^{\text{fix}})^{\tilde{\alpha}_L} K_{it}^{\tilde{\alpha}_K}$. The corner $L_{it}^{\text{var}} = 0$ is therefore a viable interior choice rather than a degenerate exit. This is the family-substitution corner that plays a central role in the heterogeneous incidence of the adjustment friction.

Productive capital K_{it} enters production but is not the focus of the counterfactual analysis. We do not consider K_{it} with its own adjustment friction in the dynamic programme below. The farm faces a variable-labour cost of $w \, L_{it}^{\text{var}}$ per period, with w the wage rate.

3.3 The firm's problem

We combine the demand schedule from Equation (3) with the production function from Equation (5) to obtain the firm's revenue. Under market clearing ($q_{it}^D = Y_{it}$) and the standard monopolistic-competition optimal price under CES demand, revenue is

$$S_{it} = A_{it} \, Y_{it}^{(\sigma-1)/\sigma}, \tag{6}$$

where $A_{it} \equiv b_{it}^{(\sigma-1)/\sigma} \, P_t^{(\sigma-1)/\sigma} \, E_t^{1/\sigma}$ is a composite revenue shifter combining the variety-specific taste b_{it}, the CES price index P_t, and aggregate wine expenditure E_t. The exponent $\theta \equiv (\sigma - 1)/\sigma \in (0, 1)$ captures the substitution response of consumers. The firm cannot expand revenue one-for-one with output, because consumers shift toward other varieties.

Substituting the production function (Equation (5)) into Equation (6) and defining revenue elasticities $\alpha_L \equiv \theta \, \tilde{\alpha}_L$ and $\alpha_K \equiv \theta \, \tilde{\alpha}_K$ (the technological elasticities discounted by the markup factor), the reduced-form revenue function becomes

$$S_{it} \;=\; A_{it}\, e^{z_i} \left(L_i^{\text{fix}} + L_{it}^{\text{var}}\right)^{\alpha_L} K_{it}^{\alpha_K}, \tag{7}$$

where, by a slight notational simplification, we let z_i denote the permanent revenue shifter, combining the markup-discounted technological productivity (θ times the productivity term in Equation (5)) and any permanent demand-side component of b_{it} (terroir reputation, brand, established commercial position). The revenue elasticities α_L and α_K are identified directly from a regression of current production value on inputs (Section 6), without separately recovering the technological elasticities $(\tilde{\alpha}_L, \tilde{\alpha}_K)$ and the demand elasticity σ. This is the standard reduced-form revenue function in firm-dynamics models with differentiated products (Cooper and Haltiwanger, 2006; Bloom, 2009).

We model the adjustment cost on variable labour as asymmetric. Expanding the workforce is costless while contracting it carries a per-worker cost. Formally,

$$C^L = c_f^L \cdot \max(L_{-1}^{\text{var}} - L^{\text{var}\prime}, 0), \tag{8}$$

where c_f^L is the per-worker contraction cost and the max operator selects only the contraction case ($L_{-1}^{\text{var}} > L^{\text{var}\prime}$). The motivation for this asymmetric form is developed in the previous subsection.

Concerning the dynamic optimisation, each year the farm inherits the previous-period variable-labour stock $L_{i,t-1}^{\text{var}}$, observes the current demand state A_{it} and capital state K_{it} given farm-specific characteristics (L_i^{fix}, z_i), and chooses next-period variable labour $L^{\text{var}\prime}$ to maximise

$$V(L_{-1}^{\text{var}}, A, K; L^{\text{fix}}, z) = \max_{L^{\text{var}\prime} \geq 0} \Big\{ \pi - C^L + \beta_d\, \mathbb{E}_t V(L^{\text{var}\prime}, A', K'; L^{\text{fix}}, z) \Big\}, \tag{9}$$

where the flow operating profit is $\pi = S - wL^{\text{var}\prime}$, with S given by Equation (7) evaluated at the chosen $L^{\text{var}\prime}$. The demand state A_{it} and the capital state K_{it} are stochastic states whose laws of motion are specified in Section 6. Under rational expectations, the firm uses these laws to form the expectation operator $\mathbb{E}_t$ in Equation (9). Capital is exogenous to the firm's annual choice (the model imposes no friction on the K margin), so the only choice variable is $L^{\text{var}\prime}$. Throughout the paper, a stock written without subscript or prime is current-period, x_{-1} denotes the previous period, x' the next-period choice, and x^* the frictionless static optimum.

3.4 Identification target

A first-order expansion of the interior policy function implied by the Bellman in Equation (9) delivers

$$\Delta \ln L_{it}^{\text{var}} = \beta_L\, a_{it} + \eta_{it}^L, \tag{10}$$

where η_{it}^L bundles the higher-order terms the expansion drops, namely the curvature of the policy function in the lagged labour state $L_{i,t-1}^{\text{var}}$, the kink at the corner $L^{\text{var}} = 0$ that a linear approximation cannot represent, and idiosyncratic measurement error. Equation (10) is therefore a local linearisation of the structural policy, valid in the neighbourhood of typical farm-years on the interior-margin subsample. The slope β_L is the population OLS projection of $\Delta \ln L_{it}^{\text{var}}$ on a_{it} on the interior-margin subsample ($L_{it}^{\text{var}} > 0$ and $L_{i,t-1}^{\text{var}} > 0$). It is the auxiliary moment that the IV in Section 4 delivers and that the indirect inference in Section 6 matches together with the share of farm-years at the corner $L^{\text{var}} = 0$.

To recover a_{it} from data, take logs of the reduced-form revenue function (Equation (7)),

$$\ln S_{it} = \ln A_{it} + z_i + \alpha_L \ln(L_i^{\text{fix}} + L_{it}^{\text{var}}) + \alpha_K \ln K_{it}. \tag{11}$$

We project $\ln S_{it}$ onto farm and year fixed effects by survey-weighted OLS. The farm fixed effect α_i absorbs the permanent revenue shifter z_i defined below Equation (7), which combines markup-discounted productivity with terroir reputation, brand, and established commercial position. The year fixed effect λ_t absorbs every component common to every farm in a given year, namely the price-index term in $\ln A_{it}$, the aggregate-expenditure term, and any common-year shift in tastes. We then define

$$a_{it} = \ln S_{it} - \hat{\alpha}_i - \hat{\lambda}_t, \tag{12}$$

standardised to unit variance. This is the standard residualisation of log revenue used to construct an unobserved firm-level demand shock in the structural firm-dynamics literature (Cooper and Haltiwanger, 2006; Cooper and Willis, 2009; Bloom, 2009). Because a_{it} is in standard-deviation units, β_L in Equation (10) is the variable-labour response to a one-standard-deviation demand shock.

The residual a_{it} captures the firm-year idiosyncratic component of $\ln A_{it}$ (the demand shock proper) together with the within-farm-within-year variation in inputs $\alpha_L \ln(L_i^{\text{fix}} + L_{it}^{\text{var}}) + \alpha_K \ln K_{it}$ that the two fixed effects do not absorb. Inputs are co-chosen with labour each period, so a_{it} correlates with L_{it}^{var}. This is why the residual is endogenous and the IV in Section 4 is needed.

A useful benchmark is the frictionless static optimum. The first-order condition on L^{var} implies that a demand shock reaches variable labour through a leverage factor

$$1 + \frac{L_i^{\text{fix}}}{L_{it}^{\text{var},*}} > 1,$$

which exceeds one because the additive permanent-labour input does not adjust, so the variable margin alone bears the response. The full frictionless slope β_L^{free} that combines this leverage with the revenue-elasticity multiplier and the standardisation of a_{it} is derived in Online Appendix F. The contraction friction c_f^L damps β_L below β_L^{free}, and a smaller $\beta_L/\beta_L^{\text{free}}$ ratio means a larger friction.

The next section develops the IV strategy that delivers an estimate of β_L.

4 Empirical strategy

We first describe the FADN panel, then develop the instrumental-variable design that identifies the labour response to a demand shock.

4.1 Data

We use farm-level data from the FADN over 2003–2024, restricted to specialised wine farms. The unfiltered panel contains 26,796 farm-year observations on 3,872 farms. After standard restrictions detailed in Appendix A and the intensive-margin filter, the headline IV sample is 13,383 farm-year observations on 1,720 farms in 13 regions. We exclude Champagne-Ardenne because California produces only still wine and the IV's weather shock cannot propagate through the trade network to sparkling-wine production. All reduced-form estimates use the FADN survey weights to target the population-representative responsiveness of variable labour.[4]

Labour margins. Total labour L_{it} is measured in annual work units (UTA). Our focal margin is variable labour, defined as $L^{\text{var}} = L - L^{\text{fix}}$, the residual after netting out permanent labour L^{fix} (family workers and full-time employees on open-ended contracts). L^{var} captures all non-permanent hired labour. It combines fixed-term contracts (typically renewed across years to retain trained workers and returning seasonal hires) and seasonal contracts for harvest and pruning. Both components give L^{var} meaningful year-to-year persistence; it is not a pure annual flow.

Capital, sales, climate, and trade. Productive capital K_{it}^{prod} aggregates machinery, permanent plantations, farm buildings, and inventory, deflated by the INSEE agricultural-investment price index. Farm sales S_{it} are total wine revenue deflated by the INSEE agricultural-output price index. Weather variation enters the analysis through the instrumental-variable design of Section 4.2 via four climate dimensions at California (the wine-growing core of the United States), namely

[4]The structural simulator is an unweighted homogeneous-farm panel; under the identifying assumption that the adjustment-cost primitive c_f^L is common across the wine-farm population, the weighted data moment and the unweighted simulator moment converge to the same structural parameter.

drought severity, spring diurnal temperature range, ripening-window mean temperature, and spring mean temperature. All four are constructed from the CRU TS 4.09 monthly climate grids of the Climatic Research Unit, University of East Anglia (Harris et al., 2020). Drought severity is the negative ratio of growing-season (April-September) precipitation to its long-run monthly mean. Higher values correspond to drier vintages. The dimension captures yield risk through water availability, a binding constraint in California's rain-fed and irrigation-managed vineyards. Spring temperature is the mean of April-May monthly temperature. It captures early-season warmth at the budburst, bloom, and crop-set stages. These stages determine the number of berries that develop. Ripening-window temperature is the mean of August-September temperature in the veraison-to-harvest period. It captures the heat regime under which sugar accumulation and acid degradation take place. Excessive heat in this window degrades both yield (heat stress) and wine quality (loss of acid balance). Spring diurnal temperature range is the mean of April-May daily maximum-minus-minimum temperature. High values signal clear-sky, continental conditions associated with frost risk at budburst and early bloom. The four dimensions span complementary biological channels through which California weather affects vineyard productivity, namely water availability, bloom and crop-set, ripening, and early-season frost risk. International wine trade flows come from CEPII's BACI bilateral trade database, complemented by 2007 French regional wine-export tabulations from French customs (Douanes Françaises). Construction details and source files appear in Appendix A.

We measure the year-to-year change in variable labour by the one-period log-difference

$$g_{it}^{\text{var}} = \ln L_{it}^{\text{var}} - \ln L_{i,t-1}^{\text{var}},$$

defined when L_{it}^{var} and $L_{i,t-1}^{\text{var}}$ are both strictly positive and the lag is the immediately preceding year, restricting the sample to the intensive-margin subsample.

4.2 Identification: U.S. wine-supply shocks routed through the 2007 export network

We estimate the structural slope β_L defined in Equation 10 by the auxiliary regression

$$g_{it}^{\text{var}} = \alpha_i + \lambda_t + \beta_L \, a_{it} + \xi \ln K_{i,t-1}^{\text{prod}} + u_{it}, \tag{13}$$

where g_{it}^{var} is the year-on-year log-difference of variable labour on the intensive-margin subsample, a_{it} is the standardised two-way fixed-effects residual of log sales (defined in Section 3.4; it bundles the structural firm-year demand shock with within-farm-within-year input variation that the two fixed effects do not absorb), and $\ln K^{\text{prod}}_{i,t-1}$ is a pre-determined scale control. The IV-identified coefficient $\hat{\beta}_L$ is the structural moment of the model.

Because inputs are co-chosen with labour each period, a_{it} is correlated with L^{var}_{it} and the OLS slope is biased by simultaneity. To isolate the demand component, we instrument a_{it} with four region-year shocks built from weather at California, the wine-growing core of the United States. The United States is both the world's largest wine importer and a major producer; when its domestic supply falls, U.S. import demand rises and the conditions facing French wine in markets where France competes shift. The four instruments are pre-determined by the 2007 composition of French regional wine exports across destination markets.

We follow the shift-exogeneity framework of Borusyak et al. (2022), where the share-base-year choice is about instrument relevance rather than predetermination. We use 2007 because it sits at the peak of the hyperglobalisation period documented by Subramanian and Kessler (2013), a quarter-century during which the world trade-to-GDP ratio rose from roughly 40% in the early 1990s to about 60% by 2007 and international integration reached unprecedented levels. By 2007 the international wine trade network had reached the composition in which the cross-sectional spread of $\Omega^0_{r,\text{USA}}$ best captures French regions' network-mediated exposure to U.S. demand. Earlier years (still adjusting to recent EU enlargement) and post-2008 years (the Great Trade Collapse and the cascade of shocks discussed in the Introduction) deliver noisier share structures and weaker first-stage power. Identification rests on the quasi-randomness of the climate shifts.

We build each instrument in two layers. For each of four climate dimensions c at California, we route the within-region z-score $z^c_{\text{USA},t}$ through an aggregator built on the 2007 wine-trade network:

$$Z^c_{r,t} = \left(\sum_{m \in \mathcal{M}} \omega^0_{r,m}\, \pi^0_{\text{USA},m} \right) \cdot z^c_{\text{USA},t}, \tag{14}$$

where $\omega^0_{r,m}$ is French region r's 2007 wine-export share to market m ($\mathcal{M}$ is the set of destination countries France exports wine to), and $\pi^0_{\text{USA},m}$ is the share of country m's 2007 wine exports going to the United States as an importer. The bracketed term $\Omega^0_{r,\text{USA}} = \sum_m \omega^0_{r,m}\, \pi^0_{\text{USA},m}$ is a pre-determined exposure weight that varies across French regions. It captures region r's position in the global wine-trade network around the United States, summed over destinations m where region r is active and where market m in turn supplies the U.S. wine market.

The four climate dimensions at California span complementary phenological windows and capture distinct biological mechanisms relevant to vine biology and U.S. wine production, namely drought severity, spring diurnal temperature range, ripening-window mean temperature, and spring mean temperature. Raw indices are z-scored within California across vintages; the first stage determines the empirical sign of each instrument.

The exposure weight has a clear economic interpretation. The IV aggregator picks up French regions whose 2007 customer base is most exposed to the U.S. wine market through trade-network linkages. When California has bad weather, the U.S. wine market absorbs additional supply globally, and the conditions facing sellers in each wine-supplying country m shift in proportion to that country's own dependence on the U.S. as a destination ($\pi^0_{\text{USA},m}$). French region r's aggregate exposure to U.S. shocks accumulates these market-specific shifts, weighted by r's 2007 customer composition ($\omega^0_{r,m}$). Regions whose customer base concentrates in countries that supply heavily to the U.S. feel U.S. shocks most strongly through this network. The IV does not commit to a single propagation mechanism. Supply diversion, wholesale price clearing, and strategic re-routing each contribute in different markets but reflect the same third-country logic, and the aggregate network exposure is measured by $\Omega^0_{r,\text{USA}}$. California weather affects French wine demand through the U.S.'s position in the global wine-trade network of markets in which French producers also sell (Anderson and van Wincoop, 2003; Head and Mayer, 2014). Because the four instruments vary at the (region, year) level, we cluster the IV standard errors at the region level.

We make two identifying assumptions.

(i) Exclusion through demand only. The four Z's affect French farm-level labour decisions only through the contemporaneous demand shock a_{it}. The assumption is plausible because California weather has no direct agronomic or labour-market link to French vineyards. The two regions share neither a growing season nor a labour pool, so the only route from a Californian climate shock to French hiring runs through the wine-demand channel the instrument is built to capture. Year fixed effects absorb any shock common to all French regions; the identifying variation is each region's differential exposure to U.S. weather through $\Omega^0_{r,\text{USA}}$.

(ii) Quasi-random shifts under the shift-share framework. Following Borusyak et al. (2022), identification rests on the as-if-random assignment of the shifts rather than on strict share predetermination. The time-varying components of our instruments are California-weather z-scores at the (vintage, climate-dimension) level. Neither French wine farms nor the global wine market influence the Pacific climate processes that generate them, so the shifts are plausibly exogenous to French wine demand by geographic and climatological reasoning. The 2007 exposure weight $\Omega^0_{r,\text{USA}}$, built from 2007 French regional wine-export shares ($\omega^0_{r,m}$) and 2007 BACI bilateral wine flows ($\pi^0_{\text{USA},m}$), is predetermined relative to the year-by-year climate variation that drives identification. Modern weak-IV diagnostics and cluster-robust inference adapted to the shift-share structure of the instrument are reported

in Online Appendix D.

5 Results

5.1 The 4-IV: reduced form and 2SLS estimates

We estimate Equation (13) by 2SLS, instrumenting a_{it} with the four climate dimensions at California routed through the 2007 trade-network exposure weight $\Omega^0_{r,\mathrm{USA}}$, on the variable-labour intensive-margin sample (the 13-region subsample of Section 4.1). Table 1 reports the resulting estimates.

	Variable labour (g_{it}^{var})
Panel A: Reduced form (OLS, farm-clustered)	
$\hat{\psi}_1$	0.0269^{**}
	(0.0109)
Panel B: 2SLS (region-clustered, 4 instruments)	
$\hat{\beta}_L$	0.3289^{***}
	(0.0654)
Panel C: First-stage diagnostics (region-clustered)	
First-stage F	37.17
First-stage F (LOO min, $k = 3$)	8.10
Hansen J p	0.604
Hansen J p (LOO min)	0.403
AR 95% CI (asymptotic-G)	$[+0.186, +0.515]$
AR 95% CI (small-G)	$[+0.140, +0.574]$
N	13,383
Farms	1,720
Regions	13

Note: Outcome $g_{it}^{\text{var}} = \ln L_{i,t}^{\text{var}} - \ln L_{i,t-1}^{\text{var}}$ on the variable-labour intensive-margin subsample ($N =$ 13,383 farm-years on 1,720 farms in 13 regions). Control: $\ln K_{i,t-1}^{\text{prod}}$. Fixed effects: farm and year. Weights: FADN survey weights. *Panel A*: OLS reduced form, farm-clustered SE. *Panel B*: 2SLS on the four-instrument stack, region-clustered SE. *Panel C* reports first-stage diagnostics: joint F on the four instruments and the leave-one-out (LOO) minimum F over the four three-instrument drops ($k = 3$); the Stock and Yogo (2005) 5%-maximal-relative-bias threshold (Table 5.1) for $k = 4$ is 16.85 and the 10%-maximal-relative-bias threshold for $k = 3$ is 9.08. The Hansen J p-value is the IVGMM over-identification test (df $= 3$; LOO df $= 2$). The Anderson–Rubin 95% CIs are from grid inversion: the asymptotic-G version against the χ_4^2 critical value, the small-G version against the Andrews et al. (2019) $F(4, G-4)$ correction for $G = 13$ region clusters. Standard errors in parentheses. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 1: Reduced form and instrumental variable identification on variable labour

The OLS reduced form in Panel A delivers a positive and significant slope. Variable-labour hiring co-moves with the demand shock proxy. But as Section 3.4 makes explicit, the empirical a_{it} is the TWFE residual of log sales and bundles the structural demand shock with the within-farm-within-year input variation $\alpha_L \ln(L_i^{\text{fix}} + L_{it}^{\text{var}}) + \alpha_K \ln K_{it}$ that the two fixed effects do not absorb. Inputs are co-chosen with labour each period, so the OLS slope on a_{it} is biased by simultaneity. We therefore turn to 2SLS, which projects a_{it} onto the four climate dimensions at California routed through the 2007 trade-network exposure weight $\Omega_{r,\text{USA}}^0$ and uses only the projected demand-side variation.

The 2SLS estimate identifies a positive but bounded labour response (Table 1, Panel B). The implied magnitude is that a one-standard-deviation increase in the demand shock raises variable labour by roughly a third of one log point at the intensive margin.

The 4-IV passes the standard weak-IV battery (Table 1, Panel C). The joint first-stage F clears the Stock and Yogo (2005) 5%-bias threshold for $k = 4$ instruments and the Anderson–Rubin weak-IV-robust 95% CI excludes zero. The leave-one-out F minimum sits below the 10%-bias threshold for $k = 3$, but the 2SLS coefficient in that worst-case subset remains positive and significant.

The Hansen J test asks a complementary question. With four instruments and one endogenous variable, the system is over-identified. Each instrument provides a separate moment that estimates the same parameter β_L. If the four climate dimensions actually identified different elasticities (for example because one instrument violated the exclusion restriction), the J test would reject. It does not reject in our sample; the four climate dimensions identify a common elasticity (Online Appendix D).

Finally, the binding-catastrophe event-study (Section 2.2) implies a contemporaneous elasticity from the ratio of its event-year labour and revenue responses that is of the same order of magnitude as the 2SLS estimate, though the event-year labour coefficient is not itself individually significant. The two designs rest on entirely different variation, cross-region weather exposure here and farm-level damage timing there, so the agreement is a cross-check rather than a second identification.

5.2 Robustness of the 4-IV

The headline 4-IV is robust on several dimensions, reported in Online Appendix D. No single observation, instrument, or region drives the estimate, and alternative climate-dimension stacks give the same sign and order of magnitude. LIML and efficient GMM lie close to 2SLS, so the estimate is not contaminated by many-weak-instrument bias. The Olea–Pflueger effective F and the Anderson–Rubin confidence interval under multiple clustering and critical-value conventions are also reported there, and the headline survives every check under the conventional inference conventions of Table 1.

6 Structural estimation and counterfactuals

We close the model and recover the cost of the friction by indirect inference. We then quantify the operating-profit loss it creates and how that loss falls across farm types.

6.1 Family substitution and heterogeneity

As already described in Section 3.3, a producer of permanent type z_i enters period t with state $(L^{\text{var}}_{i,t-1}, A_{it}, K_{it})$, observes the demand state and the capital state, and chooses next period's variable labour $L^{\text{var}}_{it} \geq 0$. The flow payoff is operating profit net of the asymmetric contraction cost defined in Equation (8). The value function satisfies the recursive Bellman equation (9), whose right-hand side contains the firm's expectation of next period's continuation value. We close the model under rational expectations (the firm computes this expectation using the true conditional distribution of next period's states given today's states). We do not model entry, exit, or general-equilibrium wage adjustment.

The model has a family-substitution corner. The permanent labour input L^{fix}_i enters the labour aggregator additively. Revenue remains positive when $L^{\text{var}}_{it} = 0$, since the permanent workforce alone delivers $A_{it}\, e^{z_i}(L^{\text{fix}}_i)^{\alpha_L} K^{\alpha_K}_{it}$. When variable-labour hire goes to zero in a low-demand year, family and full-time staff continue producing, and the corner $L^{\text{var}}_{it} = 0$ is a viable interior choice rather than a degenerate exit.[5] Re-entry from the corner is free.[6]

6.2 Calibration

The structural model has eleven parameters. Ten are calibrated outside the inference loop, leaving the contraction cost c^L_f to be estimated by indirect inference in Section 6.4.

To pin down the revenue elasticities α_L and α_K that appear in the reduced-form revenue function (Equation (7)), we estimate Equation (11) by survey-weighted OLS on FADN with farm and year fixed effects and farm-clustered standard errors. The empirical counterpart of $\ln S_{it}$ is the log of current gross production at current prices (gross output plus intra-consumption, the standard agricultural-statistics measure of farm-year production value), deflated by the INSEE agricultural-output price index. The farm fixed effect absorbs the permanent revenue shifter z_i and the

[5]The IV regression in Section 5.1 restricts to the intensive-margin subsample ($L^{\text{var}}_{i,t} > 0$ in both t and $t-1$); extensive-margin switches at zero are outside the smooth-response slope β_L but are absorbed by the corner mechanism described here, and the share of farm-years at the corner is the second moment that disciplines the indirect inference of Section 6.4.

[6]At $L^{\text{var}}_{-1} = 0$, the contraction term becomes $\max(0 - L^{\text{var}\prime}, 0) = 0$ for any $L^{\text{var}\prime} \geq 0$. The contraction cost is paid only for moves toward zero.

year fixed effect absorbs the aggregate-year component of $\ln A_{it}$, so the regression identifies the revenue elasticities from within-farm-within-year variation in log sales. By the definition $\alpha_L \equiv \theta\,\tilde{\alpha}_L$ and $\alpha_K \equiv \theta\,\tilde{\alpha}_K$ in Section 3.3, the estimates deliver the products without separately recovering the markup factor θ and the technological elasticities $(\tilde{\alpha}_L, \tilde{\alpha}_K)$.

The same regression delivers the cross-farm dispersion of the permanent revenue shifter, σ_z. The farm fixed effects are the empirical counterparts of z_i in Equation (7), so the cross-farm standard deviation of the estimated fixed effects identifies σ_z. The three estimates are $\hat{\alpha}_L = 0.32$, $\hat{\alpha}_K = 0.13$, and $\hat{\sigma}_z = 0.59$ (Table 2).[7]

We now turn to the laws of motion of the stochastic states. Rational expectations requires the firm to integrate the operator $\mathbb{E}_t$ in the Bellman against true conditional distributions of next-period states. The structural firm-dynamics literature standardly models such processes as first-order autoregressions in logs (Cooper and Haltiwanger, 2006; Cooper and Willis, 2009; Bloom, 2009), and we follow this convention. The firm-year idiosyncratic component a_{it} of $\ln A_{it}$ defined in Equation (12) is the only object on the demand side of the firm's state space that has stochastic dynamics. The permanent revenue shifter z_i from Equation (7) is constant per farm, and the aggregate-year component of $\ln A_{it}$ (the year fixed effect λ_t in Section 3.4) is treated as deterministic from the firm's perspective. The two processes are

$$a_{i,t+1} = \rho_A\, a_{it} + \varepsilon^A_{i,t+1}, \qquad \varepsilon^A \sim \mathcal{N}(0, \sigma^2_{\varepsilon,A}), \tag{15}$$

$$\ln K_{i,t+1} = \rho_K \ln K_{it} + \varepsilon^K_{i,t+1}, \qquad \varepsilon^K \sim \mathcal{N}(0, \sigma^2_{\varepsilon,K}), \tag{16}$$

with innovations drawn independently from each other.

To estimate the AR(1) parameters in Equations (15) and (16), we need empirical realisations of a_{it} and $\ln K_{it}$ for each farm-year. The capital state $\ln K_{it}$ has a direct empirical counterpart, $\ln K^{\text{prod}}_{it}$. We fit Equation (16) by survey-weighted OLS on the within-farm component of $\ln K^{\text{prod}}_{it}$ (after removing farm and year fixed effects), and recover $\hat{\rho}_K$ and $\hat{\sigma}_{\varepsilon,K}$. The idiosyncratic demand component a_{it}, by contrast, is not directly observed. We recover its empirical counterpart from the log-form revenue equation (Equation (11)). Residualising $\ln S_{it}$ on farm and year fixed effects absorbs the permanent revenue shifter z_i and the aggregate-year component of $\ln A_{it}$, and partialling out $\ln K^{\text{prod}}_{it}$ removes the within-farm contribution of capital. We fit Equation (15) by survey-weighted OLS on this residual and recover

[7] The labour and capital elasticities recovered from Equation (11) are subject to the standard simultaneity bias from contemporaneous input choices (Olley and Pakes, 1996; Ackerberg et al., 2015). The value $\hat{\alpha}_L = 0.32$ lies within the range of agricultural elasticity estimates that correct for this bias. The structural estimate $\hat{c}^L_f$ is robust to perturbations of the labour share with the wage rate recalibrated from the static-optimum moment (Online Appendix E, R3).

$\hat{\rho}_A$ and $\hat{\sigma}_{\varepsilon,A}$. Partialling out $\ln K_{it}^{\text{prod}}$ ensures that within-farm capital persistence does not leak into the demand process, so that the two innovations can be treated as independent in the simulator. The estimates are $(\hat{\rho}_A, \hat{\sigma}_{\varepsilon,A}) = (0.207, 0.294)$ and $(\hat{\rho}_K, \hat{\sigma}_{\varepsilon,K}) = (0.686, 0.337)$ (Table 2).

The remaining parameters are calibrated on simple grounds. The permanent-labour endowment $L^{\text{fix}} = 2.0$ UTA is the panel median. The wage rate $w = 0.154$ is set so that the no-friction static optimum at the central productivity realisation matches the empirical mean of total labour. The discount factor $\beta_d = 0.95$ is conventional.

Parameter	Value	Description
Production function		
α_L	0.324	Labour share; FADN panel OLS, farm+year FE, $\hat{\alpha}_L = 0.324$ (SE 0.027), $N = 23,105$
α_K	0.134	Capital share; same regression, $\hat{\alpha}_K = 0.134$ (SE 0.012)
Prices and discount		
w	0.154	Wage rate (simulator); calibrated so $L^{\text{total},*}(z = 0, A = 1, K = 1) \approx 3$ UTA
β_d	0.95	Annual discount factor (standard)
Shock processes		
ρ_A	0.207	AR(1) persistence of the demand shock $\ln A_t$; survey-weighted AR(1) on the within-farm residual of $\ln S_{it}$ after partialling out $\ln K_{it}^{\text{prod}}$, $N = 22,591$
$\sigma_{\varepsilon,A}$	0.294	AR(1) innovation SD on the K-purged residual
ρ_K	0.686	AR(1) persistence of $\ln K_{it}^{\text{prod}}$ within farm; survey-weighted, two-way FE
$\sigma_{\varepsilon,K}$	0.337	AR(1) innovation SD on $\ln K_{it}^{\text{prod}}$
σ_z	0.591	Cross-farm permanent revenue-shifter SD; cross-farm SD of farm fixed effects in production-function regression, $N = 23,105$
Numerical implementation		
N_A	11	Tauchen nodes for the demand shock A_t (Tauchen, 1986)
N_K	7	Tauchen nodes for the capital state K_t
N_z	7	Gauss–Hermite nodes for z_i; quadrature on $\mathcal{N}(0, \sigma_z^2)$
L^{var} grid	45 + zero	45 log-spaced points in $[0.20, 50]$ plus zero for the family-substitution corner

Note: Production-function shares (α_L, α_K) are panel-direct estimates from a survey-weighted, farm-and-year fixed-effect regression of log current gross production (gross output plus intra-consumption, deflated by the INSEE agricultural-output price index) on $\ln L_{it}$ and $\ln K_{it}^{\text{prod}}$ on the FADN wine-farm panel 2003–2024. The simulator uses the rounded values shown in the Value column. The demand shock A_t and the capital state K_t follow independent AR(1) processes; the A-process is identified off the within-farm residual of $\ln S_{it}$ *after* partialling out $\ln K_{it}^{\text{prod}}$ so that the persistence and innovation variance assigned to A_t are net of within-farm capital movements. Capital K_t is exogenous to the firm's choice (no c_f^K friction) but enters production through K^{α_K} and is correlated within firm. The wage rate w is calibrated to scale matching: at the central productivity realisation $(z = 0, A = 1, K = 1)$ the static optimum matches the empirical mean of total labour (~ 3 UTA). The cross-farm productivity shifter z_i is drawn once per farm from $\mathcal{N}(0, \sigma_z^2)$ and discretised with a 7-point Gauss–Hermite quadrature. For each z-node the Bellman is solved separately on the joint (L, A, K) state. Robustness to alternative calibrations of (α_L, σ_z) is reported in Online Appendix E.

Table 2: Calibration parameters

6.3 Identification of the contraction cost

The model maps the friction parameter c_f^L to the two data moments through two distinct channels. First, the slope of the policy function with respect to the standardised demand shock, $\partial L^{\mathrm{var}\prime}/\partial a$, decreases monotonically in c_f^L on the contraction margin. A costlier contraction makes the firm less responsive to negative demand realisations and, because the policy function is asymmetric, dampens the average slope of the simulated regression of variable-labour growth on a_{it}. The IV-identified slope $\hat{\beta}_L$ therefore identifies c_f^L off the average contraction response. Second, we find that the share of farm-years at the corner $L^{\mathrm{var}} = 0$ increases in c_f^L only weakly. A higher friction widens the inaction band but does not by itself push more firms to zero. The zero rate is therefore a falsification check on the calibrated heterogeneity (σ_z, L^{fix}) rather than a second handle on the friction. With two moments and one parameter the model is over-identified by one degree of freedom.

6.4 Indirect inference and structural estimates

We estimate the adjustment-cost parameter c_f^L by indirect inference (Smith, 1993; Gourieroux and Monfort, 1996). Specifically, we choose the value of c_f^L that makes the simulated panel reproduce two IV-identified data moments, the linear demand elasticity $\hat{\beta}_L$ and the share of farm-years at the corner $L^{\mathrm{var}} = 0$. The non-cost parameters of the model are calibrated outside the inference loop.

We follow standard practice for stochastic dynamic programmes (Stokey et al., 1989; Adda and Cooper, 2003). We solve the Bellman equation by value function iteration, discretising the AR(1) shocks via the Tauchen method (Tauchen, 1986) and the cross-farm permanent type via Gauss–Hermite quadrature (Judd, 1998). Because z_i is a permanent farm-level parameter rather than a stochastic state, we solve the Bellman once for each z-node and aggregate across nodes using the Gauss–Hermite weights. We then simulate a synthetic panel and compute the two data moments on it. The same shock paths are used at every candidate c_f^L (common random numbers), which is essential for the finite-difference Jacobian that enters the sandwich standard error.[8]

[8] The state space is discretised with 11 Tauchen nodes for a_{it}, 7 Tauchen nodes for $\ln K_{it}$, 7 Gauss–Hermite nodes for z_i, and a 46-point grid for L^{var} (the zero corner plus 45 log-spaced positive points between 0.20 and 50 UTA). Iteration converges to a stationary policy in fewer than 300 sweeps at every candidate c_f^L. The simulated panel has $N = 1{,}500$ farms over 30 years, with the first 5 years discarded as burn-in.

We choose c_f^L to minimise an absolute-distance criterion

$$D(c_f^L) = \left|\hat{\beta}_L - \hat{\beta}_L^{\text{sim}}(c_f^L)\right| + 2\left|\text{zero}_{\text{data}} - \text{zero}_{\text{sim}}(c_f^L)\right|. \tag{17}$$

The two-to-one weighting reflects the relative magnitudes of the two moments. We use an absolute-distance criterion rather than an SE-weighted Hansen J because the simulator's zero-rate moment carries discretisation noise from the finite L grid; an inverse-variance criterion would up-weight this noise inappropriately. The standard error on $\hat{c}_f^L$ is the sandwich formula evaluated under common random numbers (Gourieroux and Monfort, 1996); the full formula, the components of the data-moment covariance matrix, and a robustness check for finite-simulation Monte-Carlo noise are reported in Online Appendix E.

Parameter	Estimate
Adjustment-cost primitive (estimated by indirect inference)	
$\hat{c}_f^L$ (per-worker labour contraction cost)	0.0600***
sandwich SE	(0.0045)
t-ratio	+13.37
Fit (data target Ψ_d vs. simulated moment Ψ_s at optimum)	
$\hat{\beta}_L$ (data)	+0.3289
$\hat{\beta}_L^{\text{sim}}$	+0.3278
Zero-rate (data)	0.252
Zero-rate (sim)	0.294

Note: Indirect-inference estimate of the per-worker labour contraction cost c_f^L on the labour-only Bellman of Section 3, using two data moments: the IV-identified linear demand elasticity $\hat{\beta}_L$ from Table 1 Panel B and the share of farm-years at the corner $L^{\text{var}} = 0$ on the variable-labour intensive-margin subsample (FADN 2003–2024, $N = 13{,}383$ farm-years on 1,720 farms in 13 regions). Simulator setup and shock-process calibration in Table 2; criterion in Equation (17). The reported sandwich standard error is computed under common random numbers (Gourieroux and Monfort, 1996); the full formula, the components of the data-moment covariance matrix, and a finite-simulation robustness check are in Online Appendix E. Robustness to alternative state-space grids, region leave-one-out, and labour-share perturbations is also reported there. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Indirect-inference estimate of c_f^L

The point estimate of c_f^L (Table 3) is statistically distinguishable from zero at conventional levels and implies a per-worker contraction cost on the order of 40% of the wage rate (with $w = 0.154$ from Table 2). The simulator's elasticity is within rounding of the IV target. The zero-rate moment is matched less tightly. The simulator predicts that about 29% of farm-years hit zero variable labour against 25% in the data, a 4-percentage-point over-prediction of the corner. This overshoot inflates the threshold-crossing component of the welfare gain (Section 6.6; farms whose ideal labour level hovers near zero). The simulator places too many farms at the family-substitution corner. Removing the friction therefore over-counts the wage savings from dropping to zero. The above-threshold-farms component, which operates well above the corner, is not affected. The estimate is robust to dropping any single region and to perturbations of the labour share with the wage rate recalibrated from the static optimum (Online Appendix E).

6.5 Aggregate operating-profit gain

We compute the per-farm operating-profit gain by comparing two versions of the model, one at the estimated c_f^L and the other with $c_f^L = 0$. In the no-friction case we re-solve Equation (9) while keeping the calibrated shock processes, the production-function parameters, the wage rate, and the cross-farm z distribution unchanged. For each z-type we simulate the two policies on the same shock realisations, and the within-pair difference in operating profit is the per-draw gain, averaged within z and aggregated across z using the Gauss–Hermite weights matched to $\sigma_z = 0.59$. We do not solve the transition between the two regimes; the comparison is between long-run averages, not between an initial condition and a long-run target.

We read the gain as the friction's effect on the average response of operating profit, not as a recoverable deadweight loss. With this framing, removing the labour adjustment friction raises operating profit by approximately 0.9% per farm-year on average, equivalent to about €1,555 per farm-year and an annual sector-level magnitude on the order of €54 million (about 0.7% of total sector revenue).[9]

Removing the labour friction raises the share of farm-years operating without variable-labour hire by approximately twelve percentage points, indicating that the asymmetric contraction friction prevents farms from contracting all the way to the family-substitution corner in low-revenue years. Without the friction, more farms drop variable-labour hire when permanent staff alone is sufficient to produce. The friction's direct incidence is therefore on farms whose static optimum crosses the corner. They keep paying wages they would otherwise save.

[9] Model units are converted to euros via the identity $Y = \pi + wL^{\text{var}}$, with the conversion factor κ calibrated so that population-weighted simulated revenue matches the weighted mean FADN revenue of €223,000 per farm-year ($\kappa =$ €97,000 per model unit). The sector aggregate scales by the weighted population of approximately 34,600 wine farms. Discounted at 5% over a 25-year horizon, the present value is approximately €0.76 billion; treated as a perpetuity, approximately €1.08 billion.

The €54M figure is the structural extrapolation of the LATE the IV identifies. The instrument identifies the labour response of compliers, the farms whose decisions move with the California-weather-driven shifts in U.S.-market exposure, to the demand-shock channel the IV captures. The structural model maps that LATE-calibrated friction to a population-average counterfactual via the calibrated z-distribution. Two channels could shift the magnitude. First, the simulator's zero-rate over-prediction (Section 6.4) inflates the threshold-crossing component of the welfare gain. Second, the exercise is partial equilibrium. In full equilibrium, removing the friction would put upward pressure on the wage rate in high-demand years, partially offsetting the per-farm operating-profit gain. The channel decomposition is the subject of the next subsection.

6.6 Heterogeneous incidence

The aggregate 0.9% figure conceals a sharply heterogeneous distribution across farm types. The heterogeneity reflects the cross-farm dispersion of the permanent revenue shifter σ_z calibrated from the FADN panel (Section 6.2). Because σ_z is fixed outside the inference loop rather than estimated jointly with c_f^L, the per-z decomposition is conditional on the calibrated heterogeneity rather than identified from the welfare moments. Table 4 reports the per-z-type operating-profit gain in absolute and population-weighted terms.

z	Pop. weight	Per-farm gain (%)	Per-farm gain (€/yr)	Annual sector (€M)	Share (%)
-2.22	0.001	$+0.00\%$	€0	0.0	0.0%
-1.40	0.031	$+0.00\%$	€0	0.0	0.0%
-0.68	0.240	$+0.11\%$	€72	0.6	1.1%
$+0.00$	0.457	$+0.83\%$	€1,107	17.5	32.6%
$+0.68$	0.240	$+1.03\%$	€3,214	26.7	49.6%
$+1.40$	0.031	$+1.00\%$	€8,319	8.9	16.5%
$+2.22$	0.001	$+0.29\%$	€7,215	0.1	0.3%
Aggregate	1.000	$+0.87\%$	€1,555	53.8	100%

Note: Per-z operating-profit gain from removing the labour adjustment friction c_f^L, computed by paired-draws simulation as in Section 6.5. Each row is one z-quadrature node from the calibrated $\mathcal{N}(0, \sigma_z^2)$ with $\sigma_z = 0.59$; quadrature weights sum to one across rows. *Per-farm gain* is the within-z average operating-profit difference between $c_f^L = 0$ and the estimated value, in % of the with-friction baseline and in euros. *Annual sector aggregate* scales the per-farm euro gain by the row's population weight applied to the empirical population of approximately 34,600 wine farms; the right-most column sums to the €54M aggregate reported in Section 6.5.

Table 4: Per-farm-year operating-profit gain from removing the labour friction, by permanent revenue type

Three regimes appear in the table, in line with the model prediction of Section 3.

First, mechanically passive farms ($z \leq -0.7$) make up about a quarter of the population; the per-farm gain is negligible (under one percent of operating profit) and their combined contribution to the sector aggregate is well below one percent. They operate at the corner $L^{\text{var}} = 0$ regardless of the friction.

Second, threshold-crossing farms ($z \approx 0$, the median) account for about 46% of the farm population by mass, and the per-farm operating-profit gain is on the order of one percent. These are farms whose static optimum crosses the family-substitution corner across the support of a_{it}. In good years they hire, in bad years they should let variable labour drop to zero and let family labour absorb production. The contraction cost makes the swing too costly, so they are stuck paying wages they would otherwise save. The mechanism is forward-looking. Faced with persistent demand fluctuations and an asymmetric contraction cost, the firm finds it optimal to keep paying wages in a low-demand year rather than fire workers today and pay the contraction cost again when demand recovers. The friction's incidence on this group is the family-substitution-buffer-disabled channel. Aggregated, this group accounts for roughly a third of the sector operating-profit loss.

Third, above-threshold farms ($z \geq 0.7$) account for about 27% of the population, with a per-farm operating-profit gain of similar percentage magnitude to the median (on the order of one percent) but a much larger absolute euro magnitude. These are farms whose static optimum always exceeds the family-substitution threshold so they always hire variable labour. The friction binds on intensive-margin scaling. Without it they would smoothly track the demand shock; with it they under-shoot in high-demand years and over-shoot in low-demand years. The per-farm euro magnitude is large because their operations are large, so each percentage point of damped scaling translates into a substantial euro figure. Aggregated, this group accounts for roughly two-thirds of the sector operating-profit loss.

7 Conclusion

Wine represents only 3 percent of France's agricultural land but consumes 20 percent of the country's pesticides, with treatment intensity nearly three times that of the average French crop.[10] Reducing this footprint, which is the central goal of the initial *Ecophyto* plan and its successive revisions, rests on substituting labour-intensive practices for chemistry, namely mechanical weeding, manual canopy management, and the cover-crop work that organic and high-environmental-value certifications require. The friction we estimate is not itself the constraint on these tasks, but any substitution from chemistry toward labour-intensive practices would have to flow through the variable-labour margin we do study.

We use a shift-share design with four instruments, routing California weather shocks through the 2007 trade-network exposure of French regions to the US wine market. The design identifies the population-average elasticity of variable labour to a demand shock. The IV-identified response is bounded well below the frictionless static benchmark. Indirect inference on the policy slope and on the share of farm-years at the family-substitution corner places the implied per-worker contraction cost at about 40% of the wage rate. Under the calibrated dynamic programme, removing the friction would raise sector operating profit by about 0.9% per year, on the order of €54 million annually.

Our estimate is the structural extrapolation of the LATE the IV identifies, with two channels that could shift the magnitude in either direction in a fuller exercise. Four natural extensions would sharpen and enrich this magnitude. A general-equilibrium treatment with endogenous wages would let friction relaxation feed back through the variable-labour market, partially absorbing the per-farm gain into wage adjustment. A re-entry friction at the family-substitution corner (search and contracting costs we currently set to zero) would generate a richer two-friction

[10] Simonovici et al. (2019); the 3-percent / 20-percent contrast is also reported in Bonnefoy (2012).

structure on the variable-labour margin. Jointly modelling outsourcing alongside variable labour would discipline a flexible margin that the data show responds positively to demand variation. Finally, the premiumisation of French wine over the panel period (the steady reweighting toward AOC, organic, and high-environmental-value certifications that require more labour per litre) could mean that part of what we identify as a contraction cost is a conversion cost toward more labour-intensive production, including within conventional viticulture. Disentangling these two channels is a question we leave to future work.

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A Variable definitions and descriptive statistics

The unfiltered panel includes all FADN farms classified under the specialised-viticulture orientation. Two filters define the estimation sample. We drop singleton farms and farm-year pairs with a legal-form change between $t-1$ and t (which break the labour-stock continuity). The regional restriction described in Section 4.1 applies in addition.

Total labour L_{it} is the farm's annual labour input in work units (UTA), partitioned into permanent labour L^{fix} (the farm operator, family members, and full-time employees on open-ended contracts) and variable labour $L^{\text{var}} = L - L^{\text{fix}}$ (the residual, capturing all non-permanent hired labour, namely fixed-term contracts often renewed across years and seasonal contracts for harvest and pruning, a mix carrying meaningful year-to-year worker-farm continuity). Outsourcing T_{it} is total expenditure on custom-work services purchased from agricultural contractors, deflated by the annual labour-cost index. Plant-protection expenditure and expenditure on fertilizers and soil amendments are deflated with the INSEE agricultural-input price index. Productive capital K_{it}^{prod} aggregates machinery, permanent plantations (vineyards), farm buildings, and inventory, deflated with the INSEE agricultural-investment price index; land value and forest plantations are excluded. Total farm sales S_{it} are deflated with the INSEE agricultural-output price index. The idiosyncratic revenue shock a_{it} is the survey-weighted two-way (farm and year) fixed-effects residual of $\ln S_{it}$, standardised to zero mean and unit variance. Replication-level field-name documentation is in the replication package's codebook; this paper uses only the constructed variables defined here.

	All farms	
	Mean	SD
Total labour (UTA)	2.7	2.4
Permanent labour (UTA)	2.1	1.8
Variable labour (UTA)	0.6	1.1
Sales (€, deflated)	241,025	286,754
Plant protection (€, deflated)	9,263	11,022
Fertilizer + amendments (€, deflated)	4,039	7,254
Capital (€, deflated)	267,777	377,690
Outsourcing to agricultural contractors	10,079	19,328
Observations	26,796	
Farms	3,872	

Note: Means and standard deviations on the unfiltered FADN wine-farm panel, 2003–2024, survey-weighted. Labour stocks (total, permanent, variable) are in annual work units (*Unité de Travail Annuel*, UTA). Sales, plant protection, fertilizer, capital, and outsourcing are in 2010-deflated euros (sales with the INSEE agricultural-output price index; capital with the INSEE agricultural-investment price index; the remaining expenses with the INSEE agricultural-input price index). Sample sizes are the unfiltered panel (26,796 farm-years, 3,872 farms); the estimation samples on each margin are smaller after the intensive-margin and outlier filters defined in this appendix.

Table 5: Descriptive statistics (survey-weighted)

B Outsourcing as the labour-vs-workforce comparator

The introduction's work-versus-workforce framing rests on a direct comparison between in-house variable labour and outsourced contractor services. Agricultural contractors (*entreprises de travaux agricoles*) perform the same physical operations as the farm's own variable labour (pruning, harvest, treatment campaigns) but transact each engagement as a one-shot purchase rather than within a continuing employment relationship. Nguyen et al. (2022) document that six in ten French farmers now use custom-work services, up from a marginal practice in the early 2000s. How the outsourcing margin co-moves with farm activity is *a priori* ambiguous: Dupraz and Latruffe (2015) find that hired labour complements both family and contract work on French field-crop farms, while contract and family labour are substitutes; Foster and Rosenzweig (2022) and Caunedo and Kala (2021) emphasise the substitution channel by which outsourced services replace owned inputs when indivisibilities or hiring frictions bind. The wine-farm reduced form below favours the complementarity reading.

The outsourcing level (see Appendix A for the field definition) departs from the OLS-and-2SLS log-difference framework used on the labour stocks for two reasons. First, it exhibits substantial extensive-margin churn: 16.6% of farm-year observations are zero, and year-to-year transitions between zero and positive use account for approximately 11% of all consecutive-year pairs, transitions that the log-difference cannot accommodate by construction. Second, the intensive-margin log-differences are heavy-tailed (approximately 4% of observations change by more than a factor of nine year-on-year). We therefore estimate the response by Poisson maximum likelihood with two-way fixed effects (Santos Silva and Tenreyro, 2006), which is consistent under arbitrary conditional variance and handles the extensive-margin zeros directly. Outsourcing expenditure is deflated by the annual labour-cost index rather than by the agricultural-output price index, because vineyard services are overwhelmingly labour-intensive and the labour-cost deflator measures real services purchased.[11]

[11] The output-price deflator would mechanically compress real outsourcing expenditure in years with positive wine-price shocks and bias the estimated demand response.

	Outsourcing
Panel A: First stage on a_{it} (region-clustered)	
USA drought severity ($Z_{r,t-1}^{\text{dry,USA}}$)	-0.4804 (0.4826)
USA spring diurnal range ($Z_{r,t-1}^{\text{dtr,USA}}$)	0.4064 (1.3207)
USA ripening temp ($Z_{r,t-1}^{\text{rip,USA}}$)	-0.9735 (1.4959)
USA spring temp ($Z_{r,t-1}^{\text{spr,USA}}$)	-1.6254 (1.2063)
Joint F	2.55
Panel B: PPML reduced form on $E[T_{it}]$ (region-clustered)	
ψ_1 (linear, a_{it})	$+0.1074^{***}$
	(0.0067)
ψ_2 (quadratic, a_{it}^2)	$+0.0054^{**}$
	(0.0026)
N	19,129
Farms	2,256
Regions	13
Frac. zero	16.6%

Note: *Panel A* reports the linear first stage of a_{it} on the four headline instruments of Table 1 (the four climate dimensions at California routed through $\Omega^0_{r,\text{USA}}$), region-clustered, on the PPML sample. *Panel B* reports the PPML semi-elasticity of expected outsourcing expenditure with respect to a_{it}, with farm and year fixed effects and region-clustered standard errors in parentheses. Sample is the same 13-region panel as the headline IV (Section 4.1) without the $L_{i,t}^{\text{var}} > 0$ intensive-margin filter, giving $N \approx 19{,}000$ farm-years on $\sim 2{,}260$ farms in 13 regions and including the extensive-margin zeros that PPML accommodates. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Outsourcing: 4-IV first stage and PPML reduced form

The outsourcing semi-elasticity is positive at the 1% level (Panel B): a one-standard-deviation positive demand shock raises expected outsourcing expenditure by approximately eleven percent, of the same order of magnitude as the variable-labour response in the main IV table. We deliberately do not include outsourcing on the right-hand side of the variable-labour regression of Section 4.2. It is itself an outcome of the demand shock that drives labour, and adding a treatment-responsive variable as a control would introduce post-treatment bias (Angrist and Pischke, 2009). The lagged-capital control we retain is predetermined relative to the contemporaneous shock and is not subject to the same concern.

C Robustness of the catastrophe event-study

This appendix reports two exhibits on the catastrophe event-study of Section 2.2. Table 7 is the numerical companion to Figure 1 in the body. Table 8 is the robustness battery on the indemnity-amount cohort.

	$\ln L^{\text{var}}$ (1)	$\ln S$ (2)
Panel A: Event-time profile		
$h = -3$	0.0188	-0.0229
	(0.0426)	(0.0146)
$h = -2$	0.0259	-0.0233
	(0.0342)	(0.0156)
$h = -1$	0.0000	0.0000
	(0.0000)	(0.0000)
$h = +0$	-0.0248	-0.0917***
	(0.0323)	(0.0143)
$h = +1$	-0.0405	-0.0976***
	(0.0422)	(0.0175)
$h = +2$	-0.0816	-0.0852***
	(0.0524)	(0.0204)
$h = +3$	-0.1611***	-0.0789***
	(0.0547)	(0.0220)
Overall ATT	-0.0653*	-0.0896***
	(0.0341)	(0.0135)
Panel B: Wald-ratio elasticity $\hat{\theta}^h_L / \hat{\theta}^h_S$ (within-sample)		
$h = +0$	+0.27	
$h = +1$	+0.41	
$h = +2$	+0.96	
$h = +3$	+2.04	
Treated farms	918	

Note: Sun–Abraham interaction-weighted (IW) event-study estimator (Sun and Abraham, 2021) on $\ln L^{\text{var}}$ and $\ln S$, with $h = -1$ as the omitted reference period; farm-clustered standard errors in parentheses. The treated cohort is the year of largest indemnity amount within each farm, restricted to farms with maximum indemnity $\geq$ €10,000. Panel B reports the within-sample Wald-ratio elasticity $\hat{\theta}^h_L / \hat{\theta}^h_S$. Numerical companion to Figure 1 in the body. Significance: *** 1%, ** 5%, * 10%.

Table 7: Event-study on the indemnity-amount cohort

Robustness of the indemnity-amount cohort. Table 8 reports four robustness exhibits on the indemnity-amount cohort. Panel A varies the indemnity threshold across €10,000, €20,000, €30,000, and €50,000, shrinking the cohort from 918 to 272 farms. The impact-year ($h = 0$) coefficient is small and statistically insignificant at every threshold; the post-event labour response concentrates at the third horizon, where it is significantly negative at the two looser thresholds (-0.16 at €10,000, -0.15 at €20,000) and attenuates toward insignificance as the cohort shrinks, consistent with the power loss from the smaller high-indemnity samples. Panel B re-estimates with standard errors clustered at the region level rather than the farm. With only 15 region clusters, conventional cluster-robust normal approximations are anti-conservative and the reported t-statistics should be referred to a t_{G-1} distribution.[12] Under that small-G adjustment, the $h = 3$ coefficient remains significant. Panel C compares the pooled control set with a never-indemnified-only control set. The $h = 3$ response attenuates from -0.16 to -0.10 with the smaller never-only control sample, consistent with the power loss from dropping approximately 1,900 farms. Panel D reports the sum-of-squared-t joint pre-trend test on $h \in \{-3, -2\}$ under independence between event-time coefficients, which does not reject parallel trends ($\chi^2(2) = 0.77$, $p = 0.68$). The test is conservative under positive correlation between the two pre-period coefficients, the typical case for jointly estimated event-time aggregates.

[12]We use t_{G-1} critical values rather than the wild-cluster bootstrap because the number of clusters (15) is below the regime in which the bootstrap reliably controls size (MacKinnon and Webb, 2018).

	$h=-3$	$h=-2$	$h=0$	$h=+1$	$h=+2$	$h=+3$
Panel A: Indemnity-amount threshold sensitivity (cohort = year of largest indemnity within each farm)						
INDAS $\geq$ €10,000 ($N=918$)	0.019	0.026	-0.025	-0.041	-0.082	-0.161***
	(0.043)	(0.034)	(0.032)	(0.042)	(0.052)	(0.055)
INDAS $\geq$ €20,000 ($N=619$)	-0.016	-0.013	-0.005	0.022	-0.006	-0.146**
	(0.050)	(0.042)	(0.038)	(0.050)	(0.059)	(0.066)
INDAS $\geq$ €30,000 ($N=451$)	-0.049	-0.026	-0.002	0.049	0.031	-0.108
	(0.056)	(0.047)	(0.044)	(0.059)	(0.066)	(0.072)
INDAS $\geq$ €50,000 ($N=272$)	-0.042	-0.007	0.004	0.061	0.078	-0.067
	(0.068)	(0.055)	(0.055)	(0.073)	(0.089)	(0.092)
Panel B: Region-clustered SE ($G=15$ regions; significance at t_{G-1} critical values)						
REGIO-clustered	0.019	0.026	-0.025	-0.041	-0.082	-0.161***
	(0.056)	(0.051)	(0.044)	(0.058)	(0.050)	(0.054)
Panel C: Control-group robustness (broad cohort, INDAS $\geq$ €10,000, IDNUM-clustered)						
Pooled (headline) ($N_{\text{treated}}=918$)	0.019	0.026	-0.025	-0.041	-0.082	-0.161***
	(0.043)	(0.034)	(0.032)	(0.042)	(0.052)	(0.055)
Never-indemnified controls only ($N_{\text{treated}}=918$)	-0.004	0.008	-0.013	-0.022	-0.043	-0.102
	(0.048)	(0.036)	(0.035)	(0.049)	(0.061)	(0.065)
Panel D: Joint pre-trend test on $h \in \{-3,-2\}$, sum-of-squared-t under independence (IDNUM-clustered): $\chi^2(2)=0.77$, $p=0.682$						

Note: Sun–Abraham interaction-weighted (IW) event-study estimator (Sun and Abraham, 2021) on $\ln L^{\text{var}}$. Robustness exhibits on the indemnity-amount cohort of Section 2.2. Panel A varies the indemnity threshold (€10,000 to €50,000); Panel B uses region-clustered SEs with stars referred to a $t_{G-1=14}$ distribution to account for the small cluster count; Panel C compares the headline pooled control set with never-indemnified-only controls; Panel D reports a sum-of-squared-t joint pre-trend test on $h \in \{-3,-2\}$ under independence between event-time coefficients (conservative under positive correlation, which is the typical case for jointly estimated coefficients). The reference period $h=-1$ is omitted. Significance: *** 1%, ** 5%, * 10%.

Table 8: Robustness of the event-study on the indemnity-amount cohort

Online Appendix

D Robustness of the IV (Online Appendix, Not for Publication)

We report several exhibits that probe the structure of the four-instrument identification in Section 4.2: a panel-cell decomposition of the headline 2SLS estimator, the drop-one-instrument and drop-one-region leave-one-out (LOO) of the headline 4-IV stack, limited-information maximum likelihood and efficient generalised method of moments alongside 2SLS on the same stack, alternative climate-dimension stacks, a multi-cluster AR confidence interval comparison, and the cluster-robust Olea–Pflueger effective F. All exhibits use the same endogenous variable (a_{it}), the same control ($\ln K^{\text{prod}}_{i,t-1}$), the same farm and year fixed effects, the same FADN survey weights, and the same region-clustered standard errors as Table 1.

We decompose the 4-IV 2SLS estimator at the panel-cell level via a two-step exact identity. The first step is the Imbens and Angrist (1994) weighted-average representation

$$\hat{\beta}_{2SLS} = \sum_k w_k \hat{\beta}^k_{IV}, \qquad w_k = \frac{\hat{\Pi}_k Z_k' \tilde{a}}{\sum_j \hat{\Pi}_j Z_j' \tilde{a}},$$

of the joint estimator as a weighted average of just-identified single-IV estimators, with weights w_k determined by the joint first-stage projection (the $\hat{\Pi}_k$ are the joint first-stage coefficients on Z_k and $\tilde{a}$ is the demand shock residualised on the controls and fixed effects of Equation (13)). The second step decomposes each $\hat{\beta}^k_{IV}$ at the panel-cell level by writing $\alpha^k_{r,t} = (Z_k\tilde{a})|_{r,t} / \sum_{r',t'} (Z_k\tilde{a})|_{r',t'}$ for the share of $Z_k'\tilde{a}$ contributed by panel cell (r,t). Composing the two steps gives the joint cell weight on (k,r,t) as $w_k\alpha^k_{r,t}$, and Table 9 reports the resulting decomposition.[13]

[13] This is a regression-sample decomposition of the joint 2SLS estimator: it identifies which (instrument, region, year) cells of the regression sample drive each instrument's contribution to the headline estimator. Joint cell weights $w_k\alpha^k_{r,t}$ can be negative, which is mechanical for shift-share constructions and does not signal a failure of the over-identifying restrictions.

Panel A: IA94 weights and per-instrument concentration

Instrument	w_k	$\hat{\beta}^k_{IV}$	Herfindahl	% cells $\alpha < 0$
USA dry sev.	-0.012	0.148	1.361	46.2%
USA dtr spr.	0.782	0.316	0.120	41.8%
USA ripen tmp	0.019	0.717	0.464	47.5%
USA spring tmp	0.211	0.332	0.162	44.5%

$\hat{\beta}_{2SLS} = \sum_k w_k \hat{\beta}^k_{IV} = 0.3289$ (decomposition error $= 1.1e - 15$)

Panel B: Joint 4-IV cell distribution (1,196 (instrument,r,t) cells)

Joint Herfindahl on $\lvert w_k\alpha_k\rvert / \sum \lvert w_k\alpha_k\rvert$	0.0082
% cells with $w_k\alpha_k < 0$	46.9%
Top-10 share of $\sum \lvert w_k\alpha_k\rvert$	20.2%
Largest single cell share	3.72%

Panel C: Top-10 cells by $\lvert w_k\alpha_k\rvert$

Instrument	Region	Year	$\alpha^k_{r,t}$	$w_k\alpha^k_{r,t}$
USA dtr spr.	54	2022	0.150	0.117
USA dtr spr.	54	2004	0.130	0.101
USA dtr spr.	42	2007	0.094	0.073
USA dtr spr.	42	2004	0.085	0.067
USA dtr spr.	42	2015	0.067	0.053
USA dtr spr.	54	2002	-0.062	-0.048
USA dtr spr.	72	2002	0.061	0.047
USA dtr spr.	54	2023	0.057	0.044
USA dtr spr.	54	2005	-0.055	-0.043
USA dtr spr.	42	2006	0.053	0.042

Note: Two-step decomposition of the headline 4-IV 2SLS estimator. *Panel A* reports the Imbens and Angrist (1994) instrument-level weights w_k across the four climate dimensions at California and the just-identified single-IV estimators $\hat{\beta}^k_{IV}$, with the within-instrument Herfindahl on the panel-cell $\alpha^k_{r,t}$ weights. *Panel B* summarises the joint cell distribution: the Herfindahl on the normalised $\lvert w_k\alpha^k_{r,t}\rvert$ distribution, the share of total $\lvert w_k\alpha^k \rvert$ accounted for by the top-10 cells, and the largest single cell's share. *Panel C* lists the top-10 cells by $\lvert w_k\alpha^k\rvert$. Same outcome, endogenous variable, control, fixed effects, and weights as Table 1.

Table 9: Joint cell-level decomposition of the 4-IV 2SLS estimator on g^{var}

Two features of the decomposition are worth flagging. Panel B confirms that the joint 2SLS is broadly distributed across the (instrument, region, year) cells of the regression sample, so no small set of observations drives the headline. Panel A reports the IA94 instrument-level weights w_k across the four climate dimensions at California (drought severity, spring diurnal range, ripening-window temperature, spring temperature); the three temperature- and diurnal-range indices carry the bulk of the IA94 weight, while the growing-season-drought index carries only about 1%, so identification effectively rests on three instruments, with drought-severity retained because it remains a valid moment under the over-identification test.

Table 10 reports the headline 4-IV in the first row and the four 3-IV variants that drop each instrument in turn, on the same L^{var} intensive-margin sample as Table 1. Two patterns matter for the interpretation. First, the Hansen J over-identification test does not reject in any of the four drop-one rows. The four climate dimensions identify the same demand-elasticity object whichever three-instrument subset one looks at. Second, the headline 4-IV Anderson–Rubin 95% CI excludes zero under the asymptotic χ^2_k critical value, and the four instruments together deliver stronger identification than any three-instrument subset.

Specification	k	F	$\hat{\beta}$ (SE)	AR χ^2 95% CI	AR F 95% CI	J p
Headline 4-IV	4	37.17	+0.3289*** (0.0654)	$[+0.186, +0.515]$	$[+0.140, +0.574]$	0.604
Drop USA dry sev.	3	43.88	+0.3286*** (0.0678)	$[+0.201, +0.494]$	$[+0.167, +0.538]$	0.406
Drop USA dtr spr.	3	8.10	+0.4088*** (0.1381)	$[+0.176, +0.627]$	$[+0.128, +0.738]$	0.582
Drop USA ripen tmp	3	47.49	+0.3217*** (0.0658)	$[+0.142, +0.505]$	$[+0.102, +0.545]$	0.954
Drop USA spring tmp	3	48.72	+0.3203*** (0.0631)	$[+0.201, +0.493]$	$[+0.167, +0.534]$	0.403
N				13,383		
Farms				1,720		
Regions				13		

Note: The first row reproduces the headline 4-IV. Each subsequent row drops one instrument and re-estimates the resulting 3-IV on the same panel and same specification. Region-clustered standard errors in parentheses. "AR χ^2 95% CI" is the Anderson–Rubin grid inversion against the χ^2_k critical value (asymptotic-G reference). "AR F 95% CI" is the same grid inversion against the small-cluster-corrected $k\,F(k, G-k)$ critical value of Andrews et al. (2019). J p is the Hansen over-identification p-value computed via the efficient two-step IVGMM Sargan–Hansen statistic. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Drop-one-IV leave-one-out and Anderson–Rubin CI on g^{var}

We complement the drop-one-instrument exercise with a drop-one-region leave-one-out. The 4-IV joint first-stage F ranges over $[10.94, 148.19]$ across the thirteen 12-region sub-samples, every drop clears the conventional weak-IV threshold, and the χ^2_k AR 95% CI excludes zero in twelve of the thirteen drops. The headline is stable to dropping any one of the regions in the identification sample.

Table 11 reports the limited-information maximum likelihood (LIML) estimator and the two-step efficient generalised method of moments (GMM) estimator alongside 2SLS on the headline 4-IV stack. LIML is the canonical robustness against many-weak-instrument bias (Staiger and Stock, 1997; Andrews et al., 2019). Under weak-IV bias toward OLS, 2SLS would be pulled toward the OLS reduced form (Panel A of Table 1) while LIML would not. The LIML and 2SLS point estimates lie within rounding of each other, and efficient GMM is similarly close. All three estimators agree on sign, magnitude, and significance pattern.

Estimator	$\hat{\beta}$	SE (DoF-corrected)
2SLS	+0.3289***	0.0628
LIML	+0.3506***	0.0679
GMM	+0.3690***	0.0749
N	13,383	
Farms	1,720	
Regions	13	

Note: All three estimators on the 4-IV stack with outcome g_{it}^{var} on the L^{var} intensive-margin sample. Region-clustered standard errors in parentheses. Computed on the within-transformed sample (farm and year demeaned, survey-weighted); standard errors include the absorbed-fixed-effects degrees-of-freedom correction $\sqrt{(N-k)/(N-k-n_{\text{id}}-n_T+1)}$, which inflates each row's SE by approximately seven percent relative to the linearmodels default that ignores the absorbed FE. *Significance:* $^{***}\ p<0.01$, $^{**}\ p<0.05$, $^{*}\ p<0.10$.

Table 11: 2SLS, LIML, and efficient GMM on g^{var}

The headline stacks four climate dimensions at California (drought severity, spring diurnal range, ripening-window temperature, spring temperature) routed through the 2007 trade-network aggregator $\Omega^0_{r,\text{USA}}$. We report two alternative four-instrument stacks that share the drought-severity and spring-diurnal-range core with the headline but swap the other two dimensions:

- *Variant 1.* 'dry_severity + dtr_spring + harvest_pre + spring_tmn_min', replacing the ripening- and spring-temperature dimensions with harvest-month precipitation and the spring minimum temperature.
- *Variant 2.* 'dry_severity + dtr_spring + harvest_pre + pet_ripening', replacing the spring temperature with potential evapotranspiration over the ripening window.

Both variants are routed through the same $\Omega^0_{r,\text{USA}}$ aggregator at California. Table 12 reports the headline alongside the two variants.

	Headline (1)	Variant 1 (2)	Variant 2 (3)
Panel A: First stage on a_{it} (region-clustered)			
Joint F (4 IVs)	37.17	33.58	38.24
Leave-one-out F (min over 4 drops, $k = 3$ IVs)	8.10	11.70	26.90
Hansen J p (df = 3)	0.604	0.239	0.284
Leave-one-out Hansen J p (min over 4 drops)	0.403	0.188	0.188
Panel B: 2SLS (region-clustered)			
$\hat{\beta}_L$	$+0.3415^{***}$	$+0.2231^{***}$	$+0.3314^{***}$
	(0.0679)	(0.0611)	(0.0726)
95% CI	$[+0.21, +0.47]$	$[+0.10, +0.34]$	$[+0.19, +0.47]$
N	13,383	13,383	13,383
Farms	1,720	1,720	1,720
Regions	13	13	13

Note: All three columns are 4-IV stacks of climate dimensions at California, each routed through the 2007 trade-network exposure weight $\Omega^0_{r,\text{USA}}$ on the L^{var} intensive-margin sample. *Headline*: dry_severity, dtr_spring, ripening_tmp, spring_tmp. *Variant 1*: dry_severity, dtr_spring, harvest_pre, spring_tmn_min. *Variant 2*: dry_severity, dtr_spring, harvest_pre, pet_ripening. Same outcome, control ($\ln K^{\text{prod}}_{i,t-1}$), fixed effects (farm and year), weights (FADN survey weights), and clustering (region) as Table 1. Panel A reports joint first-stage diagnostics; Panel B reports the 2SLS coefficient on the variable-labour log-difference and the asymptotic 95% confidence interval $\hat{\beta}_L \pm 1.96 \cdot \text{SE}$, reported for cross-spec comparability; the headline AR weak-IV-robust CI is in Table 1, Panel C. Standard errors in parentheses. *Significance:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12: Alternative weather-IV stacks at California: headline vs. two variants

The point estimate on β_L is positive across the three constructions (Table 12, Panel B). The headline 2SLS estimate of $+0.33$ has a region-clustered standard error of 0.07, so its 95% confidence interval is approximately $[+0.20, +0.46]$ (Table 12, Panel B, "95% CI" row). Both variant point estimates ($+0.22$ and $+0.33$) sit inside that interval, so the variants are not statistically distinguishable from the headline. The dispersion of point estimates across alternative climate-dimension choices, with $\hat{\beta}_L$ ranging from $+0.22$ to $+0.33$, is within the sampling variation of the headline itself. The Hansen J over-identification test does not reject in any column. Sign, over-identification, and statistical magnitude all survive alternative choices of the four climate dimensions.

We probe the cluster-robust inference behind the headline 4-IV under multiple conventions. Table 13 reports the Anderson–Rubin 95% confidence interval for β_L under three critical-value conventions (χ^2_k, $kF(k, G-k)$, and $k(G-1)/(G-k) \cdot F(k, G-k)$) crossed with two clustering choices (region, $G = 13$, and year, $G =$

23). The chi-squared convention is the asymptotic-G reference. The $F(k, G-k)$ corrections of Andrews et al. (2019) adjust for the small number of clusters. Region clustering is the convention reported in Table 1; year clustering is the convention closer to the shock-level inference framework of Borusyak et al. (2022), in which the asymptotic regime for shift-share IVs is in the number of shifts rather than the number of region clusters.

	Region clustering ($G = 13$)	Year clustering ($G = 23$)
Critical-value convention		
χ^2_k	$[+0.183,\ +0.514]$	$[+0.068,\ +1.916]$
$kF(k, G-k)$	$[+0.140,\ +0.573]$	$[+0.021,\ +2.220]$
$k(G-1)/(G-k) \cdot F(k, G-k)$	$[+0.104,\ +0.622]$	$[-0.017,\ +2.220]$
First-stage F	37.17	3.82

Note: Anderson–Rubin grid inversion for the headline 4-IV regression of g^{var}_{it} on the demand shock a_{it} on the L^{var} intensive-margin sample ($N = 13{,}383$). The χ^2_k critical value is the asymptotic reference for cluster-robust inference; the F-scaled critical values are the small-cluster correction of Andrews et al. (2019). Region clustering is at the 13 French wine-producing regions of Table 1; year clustering is at the 23 vintages (MILEX) over 2003-2024. The first-stage joint F statistic under each clustering convention is reported in the last row. Standard errors in parentheses.

Table 13: Anderson–Rubin 95% CI for β_L under multiple clustering and critical-value conventions

Under region clustering, the AR confidence interval excludes zero across all three critical-value conventions. The chi-squared interval is $[+0.183, +0.514]$, the small-G F-simple correction $[+0.140, +0.573]$, and the strict Andrews et al. (2019) F-adjusted correction $[+0.104, +0.622]$. The conclusion that $\beta_L > 0$ is robust to small-cluster corrections at $G = 13$ regions. Under year clustering with $G = 23$, the chi-squared and F-simple intervals also exclude zero ($[+0.068, +1.916]$ and $[+0.021, +2.220]$); the strict ASS F-adjusted correction marginally widens the lower bound to -0.017. The first-stage joint F falls from 37.17 under region clustering to 3.82 under year clustering, reflecting that all 13 regions share the same California-weather shift within a year and the cluster-robust variance at the year level is correspondingly inflated.

We document the moderate pairwise correlation of the four California climate IVs after partialling out the headline farm and year fixed effects and the lagged-capital control. Table 14 reports the 4×4 correlation matrix of the residualised instruments in the headline regression sample. The maximum off-diagonal correlation is 0.55 (between spring diurnal range and spring temperature). The Hansen

J over-identification p-value reported in Table 1 ($p = 0.60$) is consistent with this moderate correlation; it does not arise from mechanical near-collinearity that would make the over-identification test uninformative.

	Drought severity	Spring DTR	Ripening temp.	Spring temp.
Drought severity	+1.000	+0.320	−0.045	+0.203
Spring DTR	+0.320	+1.000	+0.469	+0.548
Ripening temp.	−0.045	+0.469	+1.000	+0.188
Spring temp.	+0.203	+0.548	+0.188	+1.000

Note: Pearson correlation of the four California climate instruments $Z^c_{r,t-1}$ after residualisation on $\ln K^{\text{prod}}_{i,t-1}$, farm fixed effects, and year fixed effects, in the headline regression sample ($N = 13{,}383$). Residualisation matches the exact alternating-projections fixed-effects absorber used for the 2SLS estimate.

Table 14: Pairwise correlation of the four FE-residualised climate IVs

We finally report the Olea and Pflueger (2013) cluster-robust effective F statistic as a modern weak-IV diagnostic. The effective F accounts for non-spherical residuals in the cluster-robust variance and is compared to the critical values of Pflueger and Wang (2015) Table 1 for $k = 4$ instruments and one endogenous regressor at the 5% Wald test size. Under the survey-weighted convention used throughout the paper, region clustering gives an effective F of 16.05, which clears the 10%-relative-bias threshold (11.51). Year clustering gives an effective F of 9.73, between the 10% and 20% thresholds. The unweighted variants (which match the original asymptotic derivation of Pflueger and Wang (2015)) give 8.29 under region clustering and 7.42 under year clustering; both clear the 20% threshold (6.28) but neither clears the 10% threshold. The refined t-ratio cutoffs of Lee et al. (2022) would sharpen these weak-IV diagnostics further; we do not implement the LMMP correction in this revision.

E Structural robustness on c^L_f (Online Appendix, Not for Publication)

The reported sandwich SE on $\hat{c}^L_f$ in Table 3 is computed as $\widehat{V}_{c^L_f} = \left(J'\widehat{V}_\Psi^{-1}J\right)^{-1}$, where J is the 2×1 Jacobian of the simulated moments with respect to c^L_f at the optimum (forward finite difference) and $\widehat{V}_\Psi$ is the diagonal data-moment covariance matrix with the cluster-robust variance of $\hat{\beta}_L$ from Section 5.1 and the binomial variance of the zero rate. The Jacobian is evaluated under common random numbers. The same shock path is used at the optimum and at the perturbed parameter, so the simulator-induced moment variance at the simulated panel size ($1{,}500 \times 30 = 45{,}000$ farm-years) is suppressed below the cluster-robust data-moment variance that enters $\widehat{V}_\Psi$, and one simulated panel ($S = 1$) is sufficient. The reported SE inherits the data-moment uncertainty in $\widehat{V}_\Psi$, not Monte-Carlo simulator noise; the Gourieroux and Monfort (1996) finite-simulation inflation factor is $(1 + 1/S)$, so as an upper bound the SE could be inflated up to $\sqrt{2} \approx 1.41$ to absorb the residual Monte-Carlo component, leaving the t-statistic above four and the parameter clearly distinguishable from zero.

We probe the indirect-inference fit of the labour adjustment-cost parameter along three orthogonal directions: the regional composition of the IV sample, the outlier trim on the log-difference outcome, and the production-function calibration of the labour share. Table 15 reports the perturbed data targets and the corresponding structural fits. The exhibit's purpose is the *relative stability* of $\hat{c}^L_f$ across the three perturbation classes. To keep nineteen Bellman re-solves tractable, the table uses a coarser state-space discretisation than the headline pipeline of Table 3 (35-point log-spaced positive L^{var} grid with $L_{\text{HI}} = 20$ UTA and 5-point Gauss–Hermite on z, against the headline's 45-point grid with $L_{\text{HI}} = 50$ UTA and 7-point Gauss–Hermite). At the coarser grid the absolute-distance argmin lands on the same baseline value as the headline grid (Table 3); the criterion $D(c^L_f)$ is shallow in the neighbourhood of its minimum, so the coarser grid tracks the headline argmin without displacement. Across all nineteen perturbations the appendix-grid estimate stays in a tight band around the headline value; that band is the exhibit's substantive content.

The first block re-estimates the IV-identified labour elasticity on each of the thirteen 12-region subsamples, then refits c^L_f to the perturbed target. The friction stays in a tight band across the leave-one-out drops; no single region drives the estimate.

The second block re-runs the perturbation across three values of the log-difference outlier trim, $\tau \in \{\ln 6, \ln 9, \ln 12\}$. The friction is essentially constant across the three trims; the estimate is not sensitive to the trim cutoff.

The third block varies the labour share α_L across $\{0.27, 0.32, 0.37\}$. For each α_L we re-calibrate the wage rate from the static optimum at $(z, A) = (0, 1)$ so that the no-friction static optimum still matches the empirical mean labour, and re-solve the value function on the headline c^L_f grid. The friction varies smoothly with α_L but the qualitative magnitude (a contraction cost on the order of one-third of the wage) holds across the three labour-share values. The cross-farm permanent revenue-shifter dispersion σ_z and the AR(1) shock variance σ_ε are not re-estimated under the perturbed labour share; doing so is out of scope for this exhibit.

	$\hat{\beta}_L$	$\widehat{\text{zero}}$	$\hat{c}_f^L$	D
Baseline (App. E grid)	+0.3289	0.252	0.060	0.031
R1. Region leave-one-out				
Drop REGIO=22	+0.3265	0.253	0.060	0.031
Drop REGIO=24	+0.3271	0.251	0.060	0.034
Drop REGIO=26	+0.3107	0.261	0.060	0.031
Drop REGIO=42	+0.2442	0.262	0.080	0.048
Drop REGIO=43	+0.3256	0.253	0.060	0.031
Drop REGIO=52	+0.2734	0.257	0.080	0.044
Drop REGIO=54	+0.1435	0.240	0.120	0.022
Drop REGIO=72	+0.3308	0.232	0.060	0.068
Drop REGIO=73	+0.3766	0.252	0.060	0.064
Drop REGIO=82	+0.3638	0.264	0.060	0.028
Drop REGIO=91	+0.3241	0.238	0.060	0.063
Drop REGIO=93	+0.3414	0.255	0.060	0.022
Drop REGIO=94	+0.3360	0.251	0.060	0.025
R2. Trim sensitivity				
$\tau = \ln 6$	+0.2833	0.252	0.080	0.043
$\tau = \ln 9$ (headline)	+0.3021	0.252	0.060	0.057
$\tau = \ln 12$	+0.3006	0.252	0.060	0.059
R3. Production-share sensitivity (target fixed at headline; w re-calibrated)				
$\alpha_L = 0.27$	+0.3289	0.252	0.040	0.072
$\alpha_L = 0.32$	+0.3289	0.252	0.060	0.027
$\alpha_L = 0.37$	+0.3289	0.252	0.080	0.013

Note: Each row reports a perturbed data target $(\hat{\beta}_L, \widehat{\text{zero}})$ and the corresponding structural fit $\hat{c}_f^L$ that minimises the absolute-distance criterion $|d_L| + 2|d_{\text{zero}}|$ on the simulation grid (Equation 17). R1 drops one of the thirteen IV-sample regions and recomputes the target on the remaining 12-region subsample. R2 varies the log-difference outlier trim. R3 varies the labour share with the wage rate re-calibrated from the static optimum so that the no-friction static optimum still matches the empirical mean labour; σ_z and σ_ε are not re-estimated.

Table 15: Structural robustness of $\hat{c}_f^L$ to region LOO, trim, and labour-share perturbations

F Frictionless-limit benchmark for β_L (Online Appendix, Not for Publication)

Section 3.4 introduces the policy-function slope β_L and its frictionless counterpart β_L^{free}. The friction-damped slope has no closed form under the asymmetric contraction cost and is recovered numerically by indirect inference (Section 6.4); the frictionless slope does have one. We derive it below, then explain why Section 5.1 benchmarks $\hat{\beta}_L$ against a simulator-based rather than the analytical frictionless slope.

The frictionless limit is static. At $c_f^L = 0$ the contraction term in the Bellman (Equation (9)) vanishes, so the previous-period stock L_{-1}^{var} drops out of the problem. The farm re-optimises revenue (Equation (7)) net of variable-labour cost each year, solving

$$L_{it}^{\text{var}*} = \arg\max_{L^{\text{var}} \geq 0} \; A_{it}\, e^{z_i} \left(L_i^{\text{fix}} + L^{\text{var}}\right)^{\alpha_L} K_{it}^{\alpha_K} - w\, L^{\text{var}}.$$

The static first-order condition equates the marginal revenue product of labour to the wage and solves for total labour, $L_i^{\text{fix}} + L_{it}^{\text{var}*} = \left(\alpha_L\, A_{it}\, e^{z_i}\, K_{it}^{\alpha_K} / w\right)^{1/(1-\alpha_L)}$. Differentiating in logs with respect to the composite revenue shifter $\ln A_{it}$, and using that L_i^{fix} is fixed so the variable margin absorbs the entire adjustment,

$$\frac{\partial \ln L_{it}^{\text{var}*}}{\partial \ln A_{it}} = \left(1 + \frac{L_i^{\text{fix}}}{L_{it}^{\text{var}*}}\right) \cdot \frac{1}{1-\alpha_L}. \tag{18}$$

Two forces lift the response. The Cobb–Douglas curvature multiplier $1/(1-\alpha_L)$, where α_L is the revenue elasticity defined in Section 3.3, is the standard diminishing-returns term. A higher A raises the marginal revenue product of labour, and total labour rises until the first-order condition re-equilibrates. The leverage factor $1 + L^{\text{fix}}/L^{\text{var}*}$ is specific to our setup. Because permanent labour does not adjust, the variable component alone bears any proportional change in total labour.

The auxiliary regression (Equation (13)) projects the year-on-year difference $\Delta \ln L^{\text{var}}$ on the standardised demand residual a_{it}, not the level $\ln L^{\text{var}}$ on $\ln A$. Two conversions bridge the gap. First, differencing an AR(1) shock of persistence ρ scales the level elasticity by $(1-\rho)$. Second, the empirical a_{it} has unit variance by construction (Equation (12)), while the firm-year idiosyncratic component of $\ln A_{it}$ has standard deviation $\sigma_{\ln A} = \sigma_{\varepsilon,A}/\sqrt{1-\rho_A^2}$, so the level elasticity rescales by $\sigma_{\ln A}$ when expressed against the standardised regressor. Combining the two,

$$\beta_L^{\text{free}} = \underbrace{\left(1 + \frac{L^{\text{fix}}}{L^{\text{var}*}}\right) \cdot \frac{1}{1-\alpha_L}}_{\text{level elasticity}} \cdot \underbrace{(1-\rho)\,\sigma_{\ln A}}_{\text{difference} \times \text{standardisation}}. \tag{19}$$

With the calibrated values $\alpha_L = 0.32$, $\rho_A = 0.21$, $\sigma_{\ln A} = 0.30$ (Table 2; $\sigma_{\ln A} = \sigma_{\varepsilon,A}/\sqrt{1-\rho_A^2}$), $L^{\text{fix}} = 2.0$ UTA and $L^{\text{var}*} = 1.0$ UTA at the central calibration point, Equation (19) gives $\beta_L^{\text{free}} \approx 1.05$.

The analytical value is evaluated at the central farm. The IV estimates β_L on the intensive-margin sample ($L^{\text{var}} > 0$ in both t and $t-1$), which over-samples high-z farms operating well above the family-substitution corner, where the leverage factor $1 + L^{\text{fix}}/L^{\text{var}*}$ is much smaller. Re-solving the Bellman at $c_f^L = 0$ and running the same auxiliary regression on the same intensive-margin filter yields a frictionless-simulator slope of approximately 0.64. Constructed on the same sample as $\hat{\beta}_L$, that simulator slope is the directly comparable benchmark, and the estimated friction-damped slope is roughly half of it. The labour adjustment friction halves the policy-function slope relative to the frictionless static optimum.